Faculty of Artificial Intelligence Engineering
Department of Robotics and Intelligent
Systems Engineering

SMART HOME SECURITY SYSTEM

Graduation Project II Report

Submitted in Partial Fulfillment of the Requirements for the Bachelor's Degree in Control Engineering and Robotics

Prepared by

Batoul Maher Al Hayek

Supervised by

Dr. Adel Al-Kafri

Eng. Alhussain Tofailea.

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Abstract

This project is about building a smart home security system using the Raspberry Pi 5. The main idea is to detect movement inside the monitored area and use a

camera to recognize people using a trained YOLO model. A PIR sensor is used to detect motion, and an OLED screen displays the system status. The system also includes a buzzer and additional sensors for home monitoring.

The system is programmed in Python, and its hardware and software components are connected to work together. A Telegram bot is also included in the project design to allow the homeowner to monitor and interact with the system using a mobile phone. This provides a way to access the system remotely without being physically next to the device.

The project involves testing the main functions, including motion detection, person recognition, sensor operation, and communication with the Telegram bot. These tests help identify which parts are working correctly and which parts still need improvement. The system can also be developed further by improving its recognition performance and adding more functions in the future.

Keywords

Keywords: Smart Home Security, Raspberry Pi 5, YOLO, Computer Vision, Motion Detection, Face Recognition, Internet of Things.

الملخص

يهدف هذا المشروع إلى تصميم وتنفيذ نظام أمان لمنزل ذكي باستخدام الحاسوب المصغر Raspberry Pi 5. يعتمد النظام على حساس الحركة PIR وكاميرا لالتقاط الصور عند اكتشاف الحركة، ثم استخدام نموذج YOLO المدرّب للتعرّف على الأشخاص. كما يتضمن النظام شاشة OLED لعرض حالة التشغيل، وجرس إنذار، بالإضافة إلى حساسات أخرى لدعم مراقبة المنزل.

تم تطوير البرنامج باستخدام لغة Python، مع ربط المكونات الإلكترونية والبرمجية لتعمل ضمن نظام واحد. ويتضمن المشروع أيضًا بوتًا على تطبيق Telegram، يهدف إلى إتاحة مراقبة النظام والتفاعل معه عن بُعد باستخدام الهاتف المحمول، مما يوفر وسيلة للتواصل مع النظام دون الحاجة إلى الوقوف بجانب الجهاز.

يركز المشروع على دمج هذه المكونات واختبار وظائفها الأساسية، بما يشمل كشف الحركة والتعرّف على الأشخاص وتشغيل الحساسات والتواصل مع بوت Telegram. وتساعد الاختبارات على تحديد الوظائف التي تعمل بالشكل المطلوب والجوانب التي تحتاج إلى تحسين. ويمكن تطوير النظام مستقبلًا من خلال تحسين أداء التعرّف وإضافة وظائف جديدة وفقًا النتائج الاختبارات.

الكلمات المفتاحية: المنزل الذكي، Raspberry Pi 5، YOLO، الرؤية الحاسوبية، كشف الحركة، التعرّف على الأشخاص، بوت Telegram، المراقبة عن بُعد.

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Chapter 1

General Introduction

1.1 Project Overview and Background

A smart home security system can help monitor a home and provide information about events that happen inside it. In this project, the Raspberry Pi 5 is used as the main processing unit to connect the hardware components and run the security program.

The system uses a PIR motion sensor to detect movement and an iPhone camera to capture live video. The video stream is sent to the Raspberry Pi 5 through an RTSP connection, where the trained YOLO model is used to recognize four known people. An OLED screen displays the system status, while a buzzer can be used to provide an audible alert. Additional sensors, including gas and temperature sensors, are also included to support home monitoring.

The system is programmed in Python. A Telegram bot is also part of the project design, with the aim of allowing the homeowner to monitor and interact with the system remotely through a mobile phone. The project brings together these components in one system, and its functions need to be tested to evaluate their operation and identify areas that require improvement.

1.2 Problem Statement and Significance

Home security is important because unexpected movement or the presence of an unknown person may create a security risk. A system that only detects movement cannot determine who is present, so additional processing is needed to recognize people and provide more useful information about the event.

The main problem addressed in this project is how to combine motion detection, person recognition, and other monitoring components using a Raspberry Pi 5 in one system. The system uses a PIR sensor to detect movement and a camera with a trained YOLO model to recognize four known people. A screen and buzzer are included to provide status information and alerts, while a Telegram bot is intended to support remote monitoring and interaction through a mobile phone.

The importance of this project lies in bringing these functions together in a practical system. It also provides an opportunity to test how the hardware and software work together, identify technical limitations, and determine which parts need further development.

1.3 Project Objectives

The main objective of this project is to design and implement a smart home security system using the Raspberry Pi 5. The system combines motion detection, person recognition, and additional sensors to support home monitoring.

The specific objectives of the project are:

1. To use a PIR sensor to detect movement in the monitored area.
2. To connect an iPhone camera to the Raspberry Pi 5 through an RTSP stream to provide live video for person recognition.
3. To use a trained YOLO model to recognize four known people.
4. To display the system status on an OLED screen.
5. To include a buzzer to provide an audible alert when required.
6. To integrate gas and temperature sensors for additional home monitoring.
7. To develop the system using Python and connect its hardware and software components.
8. To include a Telegram bot for remote monitoring and interaction through a mobile phone.
9. To test the main system functions and identify areas that need improvement.

1.4 Definitions and Key Concepts

This section explains the main terms used in the smart home security system.

Smart Home Security System: A system that combines electronic devices and software to monitor a home and support its security.

Raspberry Pi 5: A small single-board computer used as the main processing unit in this project. It runs the Python program and connects the system components.

PIR Sensor: A motion sensor that detects changes in infrared radiation, usually caused by the movement of people in its detection area.

YOLO (You Only Look Once): An object detection model used in computer vision. In this project, a trained YOLO model is used to recognize four known people.

Person Recognition: The process of analyzing camera images to determine whether a detected person matches one of the people represented in the trained model.

OLED Display: A small electronic screen used to show messages about the system status.

Buzzer: An electronic component that produces sound and can be used to provide an audible alert.

Internet of Things (IoT): A concept in which physical devices connect and exchange information through a network.

Telegram Bot: A software application that works through Telegram and is intended in this project to support remote monitoring and interaction with the system.

Python: A programming language used to develop the software that controls and connects the system components.

1.5 Project Scope and Technical Requirements

The scope of this project is to build a smart home security system using the Raspberry Pi 5. The system combines a motion sensor, an iPhone camera, a person recognition model, a display, an audible alarm, and additional sensors for home monitoring.

The main functions included in the project are motion detection using a PIR sensor, person recognition using a trained YOLO model, displaying system messages on an OLED screen, and providing sound alerts through a buzzer. Gas and temperature sensors are also included as additional monitoring components. A Telegram bot is part of the project design to support remote monitoring and interaction through a mobile phone.

The technical requirements include the following hardware and software components:

- Processing unit: Raspberry Pi 5.

- Camera: An iPhone camera that provides a live video stream to the Raspberry Pi 5 through an RTSP connection for person recognition.
- Motion sensor: A PIR sensor connected to the Raspberry Pi 5.
- Recognition model: A trained YOLO model for recognizing four known people.
- Display: An OLED screen for showing system status.
- Audible alert: A buzzer for sound notifications.
- Additional sensors: Gas and temperature sensors.
- Programming language: Python.
- Communication platform: A Telegram bot intended to support remote monitoring and interaction.

The system requires suitable connections between the hardware components, the necessary software libraries, and a working RTSP video connection from the iPhone camera. Its performance depends on correct hardware setup, software configuration, and testing under different operating conditions. The scope of this project focuses on integrating these components and evaluating their main functions.

1.5.1 Technical Requirements

The technical requirements of the system are based on the hardware and software components selected for the project. Each component has a specific function, and its operation should be checked during system testing. Table 1.1 summarizes the main components, their functions, and the methods used to verify their operation.

Component	Function / Requirement	Verification Method
Raspberry Pi 5	Main processing unit for running the system software.	Check that the program runs on the device.
Mobile Phone Camera (iPhone)	Captures live video and streams it to the Raspberry Pi through an RTSP connection.	Check the RTSP connection and verify that the Raspberry Pi receives the live video stream.

PIR Motion Sensor	Detects movement in the monitored area.	Test the sensor response to movement.
YOLO Model	Recognizes four known people represented in the trained model.	Test recognition using images or camera frames of the four people.
OLED Display	Displays system status and messages.	Check that the expected messages appear.
Buzzer	Provides an audible alert when triggered by the program.	Test sound output when the alarm is activated.
Gas Sensor	Provides readings related to gas detection.	Check sensor readings and response under suitable test conditions.
Temperature Sensor	Measures temperature for home monitoring.	Compare readings with a suitable reference thermometer.
Telegram Bot	Intended to support remote monitoring and interaction.	Test bot responses and message delivery.

1.6 Project Contribution and Novelty

The main contribution of this project is the integration of person recognition and home monitoring functions using the Raspberry Pi 5. The system combines a PIR motion sensor with a camera and a trained YOLO model to detect movement and recognize four known people.

Another contribution is bringing several hardware components together, including an OLED display, a buzzer, a gas sensor, and a temperature sensor. These components provide different types of information and alerts within one system. The project also includes a Telegram bot in its design to support remote monitoring and interaction through a mobile phone.

The main value of the project is the practical implementation and testing of these connected components. Rather than treating each function separately, the project aims to bring them together in a single system. Further testing and development are needed to evaluate the recognition performance, improve reliability, and confirm the operation of all functions under different conditions.

1.7 Design Methodology

The project was developed through several stages, starting with identifying the required functions and selecting suitable hardware and software components. The Raspberry Pi 5 was selected as the main processing unit because it can run the Python program and connect to the different system components.

The next stage involved preparing the hardware connections, including the PIR motion sensor, OLED display, buzzer, gas sensor, and temperature sensor. An iPhone camera was configured to provide a live video stream to the Raspberry Pi 5 through an RTSP connection. The video stream is used as the input for the trained YOLO model to recognize four known people.

After preparing the main components, the software was developed in Python to manage motion detection, camera operation, person recognition, and system messages. The project also includes a Telegram bot intended to support remote monitoring and interaction.

The final stage focuses on testing the system functions and checking how the hardware and software work together. The results of these tests are used to identify problems and determine which parts require further improvement. The system is developed step by step so that each component can be checked before evaluating the complete system.

1.8 Report Structure

This report is organized into five chapters. Each chapter covers a different part of the project, from introducing the main idea to presenting the implementation and final conclusions.

Chapter 1 – General Introduction: Introduces the project, explains the problem and its importance, presents the objectives, defines the main concepts, and describes the project scope and methodology.

Chapter 2 – Review of Literature and Previous Studies: Presents previous studies and related systems, discusses their main features, and identifies the research gap addressed by this project.

Chapter 3 – Project Analysis and Design: Describes the proposed system architecture, hardware and software components, and the main design decisions.

Chapter 4 – Implementation and Testing: Explains how the system was implemented and presents the tests performed, their results, and the limitations identified during testing.

Chapter 5 – Conclusion and Recommendations: Summarizes the project achievements, discusses the challenges encountered, and suggests possible improvements and future work.

1.9 Chapter Conclusion

This chapter introduced the Smart Home Security System and explained its main purpose, objectives, scope, and technical requirements. The project uses the Raspberry Pi 5 to connect motion detection, person recognition, a display, an audible alarm, and additional sensors within one system.

The chapter also described the main technologies used and the methodology followed during development. The system brings together hardware and software components, with the aim of supporting home monitoring and remote interaction through a Telegram bot. Testing is an important part of the project because it helps evaluate the operation of each component and identify areas that need improvement.

The next chapter reviews previous studies and related systems to understand the approaches used in this field and identify the contribution of the proposed project.

Chapter 2

Review of Literature and Previous Studies

2.1 Previous Studies

Several studies have investigated smart home security using different hardware platforms, sensors, image processing methods, and communication technologies. The following studies are relevant to the development of the proposed Smart Home Security System.

Study 1: Ubiquitous Smart Home System Using Android Application (Kumar, 2014)

Kumar presented a low-cost smart home system based on an Android application and an Arduino Ethernet board. The system included devices such as light switches, power plugs, temperature and humidity sensors, current sensors, intrusion detection sensors, gas or smoke sensors, and sirens. The application supported home control and environmental monitoring. When an intrusion was detected, the system could generate an email and activate the siren. This study is relevant to the proposed project because it demonstrates the integration of home monitoring components and alert mechanisms within one system.

Study 2: Design of Smart Home Security System using Object Recognition and PIR Sensor (Surantha and Wicaksono, 2018)

Surantha and Wicaksono developed a home security system using a Raspberry Pi 3, an Arduino board, a webcam, and a PIR motion sensor. When the PIR sensor detected movement, the camera was activated to capture an image. The system then used Histogram of Oriented Gradients (HOG) and a Support Vector Machine (SVM) to detect a human in the image. If a suspicious person was detected, an alarm was activated. The authors reported an average detection time of approximately two seconds and a detection accuracy of 89% in their evaluation. This study is relevant to the proposed project because it combines motion sensing with image processing to support home security.

Study 3: Smart Home Security System using IoT, Face Recognition and Raspberry Pi (Dhobale et al., 2020)

Dhobale and colleagues proposed an IoT-based home security system using a Raspberry Pi, a camera, a PIR sensor, and other components. The system captured an image of a person at the door and used Local Binary Patterns (LBP) for face recognition. When the person matched a known household member, the door

could be unlocked. If the person was not recognized, the system could prevent access and send an email containing the person's image to the homeowner. The design also included a door-lock mechanism and remote monitoring functions. This study is relevant to the proposed project because it applies person recognition to home security, although it uses a different recognition method and focuses on controlling access through a door.

Study 4: A Multimodal IoT-Based Home Intrusion Detection System (Bhuckory and Pudaruth, 2023)

Bhuckory and Pudaruth presented an IoT-based intrusion detection system implemented using a Raspberry Pi 4, an IP camera, a microphone, and sound and motion sensors. The system could detect people through movement, voice, or knocking. It also included person counting, face recognition, recognition of masked faces, and speaker identification. A servo motor was used as a physical door lock for authorized users. When a potential intrusion was detected, video footage and event information could be sent to cloud services, and the homeowner could receive notifications and review events through a web application. This study is relevant to the proposed project because it demonstrates the integration of several detection methods and remote communication within one home security system.

In summary, these studies demonstrate different ways to combine sensors, image processing, recognition methods, alarms, and remote communication. They provide a basis for examining how the proposed Smart Home Security System integrates a Raspberry Pi 5, an iPhone camera streaming through RTSP, a trained YOLO model for four known people, and additional home monitoring components.

2.2 Comparative Analysis of Existing Systems

This section compares the selected smart home systems in terms of their processing platforms, detection methods, sensors, alert mechanisms, and remote monitoring capabilities. The comparison helps identify the main features of each system and explains how the proposed Smart Home Security System combines person recognition with additional home monitoring functions.

System	Processing Platform	Detection / Recognition	Sensors and Alerts	Remote Monitoring
Kumar (2014) [1]	Arduino Ethernet	Intrusion detection and environmental monitoring	Environmental and intrusion sensors; siren	Android application and email alerts
Surantha and Wicaksono (2018) [2]	Raspberry Pi 3 and Arduino	Human detection using HOG and SVM	PIR sensor and alarm	Remote notification is not established in the study summary used here
Dhobale et al. (2020) [3]	Raspberry Pi	Face recognition using LBP	PIR and ultrasonic sensors; door-lock mechanism and email notification	Remote monitoring and email notification
Bhuckory and Pudaruth (2023) [4]	Raspberry Pi 4	Person detection, face recognition, masked-face recognition, and speaker identification	Motion and sound sensors; microphone, speaker, and servo door lock	Cloud-based event information, notifications, and web application
Proposed Smart Home Security System	Raspberry Pi 5	Trained YOLO model for four known people	PIR, OLED display, buzzer, gas sensor, and temperature sensor	Telegram bot planned for remote monitoring and interaction; operation requires verification through testing

The comparison shows that the reviewed systems use different approaches to home security. Some focus on home automation and environmental monitoring, while others combine motion detection with human detection or face recognition. The proposed Smart Home Security System uses a Raspberry Pi 5 and a trained

YOLO model to recognize four known people from a live video stream provided by an iPhone camera through RTSP. It also includes a PIR sensor, an OLED display, a buzzer, and gas and temperature sensors. A Telegram bot is included in the design for remote monitoring and interaction. The actual operation and performance of the integrated system must be evaluated through testing.

2.3 Research Gap and Proposed Innovation

The reviewed studies demonstrate different approaches to smart home security, including motion detection, environmental monitoring, human detection, face recognition, and remote communication. However, the selected studies differ in the hardware platforms and methods they use, and they do not all combine the same functions within one system.

The proposed Smart Home Security System aims to integrate several functions using the Raspberry Pi 5. These include motion detection through a PIR sensor, recognition of four known people using a trained YOLO model, and live video input from an iPhone camera through an RTSP connection. The system also includes an OLED display, a buzzer, a gas sensor, and a temperature sensor.

The main contribution of this project is the integration of these components within one system and the practical evaluation of their operation. A Telegram bot is also included in the design to support remote monitoring and interaction. The performance of the system, including recognition reliability and the response of the connected components, must be evaluated through testing.

Therefore, the project focuses on implementing and testing an integrated smart home security system. The results of the testing will help identify the system's limitations and determine possible improvements for future development.

2.4 Theoretical Framework

The Smart Home Security System is based on several hardware and software technologies that work together to support home monitoring and security.

Raspberry Pi 5

The Raspberry Pi 5 is a single-board computer used as the main processing unit in the proposed system. It runs the Python program and connects the different hardware and software components. It receives video from the iPhone camera and processes the video frames for person recognition.

PIR Motion Sensor

A Passive Infrared (PIR) sensor detects changes in infrared radiation within its detection area. These changes can indicate the movement of a person. In this project, the PIR sensor is used to trigger the security program when movement is detected.

YOLO Object Detection Model

YOLO stands for "You Only Look Once." It is a family of object detection models designed to process images and locate objects efficiently. In this project, a trained YOLO model is used to classify detected people into four trained identity classes. Its performance depends on the quality of the training data, the model configuration, and the conditions in which the camera operates.

iPhone Camera and RTSP Streaming

The iPhone camera provides live video for the system. Real-Time Streaming Protocol (RTSP) is used to transmit the video stream over the network to the Raspberry Pi 5. The received frames provide the visual input for the recognition process. The reliability of this function depends on the network connection and the availability of the video stream.

OLED Display and Buzzer

An Organic Light-Emitting Diode (OLED) display is used to show short status messages, such as system readiness and processing status. A buzzer produces an audible signal when activated by the program. These components provide local information and alerts to the user.

Gas and Temperature Sensors

The gas sensor is intended to detect the presence of certain gases, depending on its type and calibration. The temperature sensor measures the surrounding temperature. These sensors extend the system beyond person recognition by providing additional environmental monitoring functions.

Python Programming Language

Python is used to develop the software that coordinates the system components. It supports the implementation of motion detection, video processing, model inference, sensor reading, and status messages through suitable libraries.

Telegram Bot and Remote Communication

A Telegram bot can provide a communication interface between the system and the user. In the proposed design, it is intended to support remote monitoring and interaction through a mobile phone. Its actual operation depends on correct configuration, network access, and successful testing.

In summary, these technologies provide the theoretical basis for integrating motion detection, person recognition, environmental sensing, and user notifications in the proposed system. Their practical performance is evaluated during the implementation and testing stage.

2.5 Chapter Conclusion

This chapter reviewed previous studies related to smart home security systems and discussed the main technologies used in this field. The reviewed systems demonstrated different approaches to motion detection, person recognition, environmental monitoring, and remote communication.

The comparison highlighted the differences between these systems in terms of their processing platforms, detection methods, sensors, and notification mechanisms. Based on this review, the proposed Smart Home Security System aims to integrate a Raspberry Pi 5, an iPhone camera connected through an RTSP stream, a trained YOLO model for recognizing four known people, a PIR

motion sensor, an OLED display, a buzzer, and additional gas and temperature sensors.

The chapter also presented the theoretical concepts behind the main hardware and software components. The next chapter describes the analysis and design of the proposed system, including its architecture and the connections between its main components.